

THE HYPER-COHOMOLOGICAL ENTROPY MANIFOLD (HCEM) THEOREM: A NON-COMMUTATIVE ALGEBRAIC-TOPOLOGICAL FRAMEWORK FOR POST-QUANTUM CRYPTOGRAPHY

MUHAMMAD AIDIL AMRY

ABSTRACT. Current post-quantum cryptographic primitives depend fundamentally on the properties of commutative rings and lattices in Euclidean space. We introduce the Hyper-Cohomological Entropy Manifold (HCEM) Theorem, providing a framework that binds discrete non-commutative algebraic structures to continuous topological invariants. By defining the Non-Abelian De Rham Cohomology space with coefficients in a C^* -algebra, we establish an asymptotic growth rate of the trace of the heat kernel. We prove that mapping the continuous entropy signature back to the discrete generator requires searching an exponentially dense phase space with zero global periodicity, rendering Quantum Fourier Transform (QFT) based period-finding algorithms mathematically void.

1. INTRODUCTION

The epistemological gap in modern cryptographic topology lies in the reliance on commutative structures. While resistant to standard applications of Shor's Algorithm, primitives such as Learning With Errors (LWE) and NTRU remain theoretically vulnerable to advanced quantum sub-lattice reduction attacks if periodicity can be exploited in higher dimensions. Previous attempts to utilize non-commutative structures, such as Braid Group cryptography, have collapsed under heuristic algebraic attacks, specifically conjugacy search problems.

This paper formally resolves this anomaly by establishing a mathematical framework that merges continuous analytical tools of Atiyah-Singer Index Theory with the discrete hardness of word-hyperbolic group theory.

2. MATHEMATICAL FORMULATION

Let $\mathcal{A}$ be a non-commutative C^* -algebra, and Λ be a discrete non-abelian subgroup acting properly discontinuously on a Riemannian manifold $\mathcal{M}$ of dimension n , where $n \geq 11$.

Definition 2.1 (Non-Abelian De Rham Cohomology). *We define the Non-Abelian De Rham Cohomology space with coefficients in $\mathcal{A}$, denoted $H_{dR}^*(\mathcal{M}; \mathcal{A})$.*

Definition 2.2 (Hyper-Cohomological Entropy). *The Hyper-Cohomological Entropy, S_τ , of the system is defined by the asymptotic growth rate of the trace of the heat kernel acting on the space of differential forms $\Omega^k(\mathcal{M}; \mathcal{A})$:*

$$(1) \quad S_\tau(\mathcal{A}, \Lambda, \mathcal{M}) = \limsup_{t \rightarrow 0^+} \left(-t \log \text{Tr}_{\mathcal{A}} \left(e^{-t\Delta_k} \right) \right) \otimes \chi(\mathcal{M}/\Lambda)$$

where $\Delta_k = dd^* + d^*d$ is the Hodge-Laplace operator twisted by the representation of the fundamental group $\pi_1(\mathcal{M}) \cong \Lambda$ into $\mathcal{A}$, $\text{Tr}_{\mathcal{A}}$ is the canonical von Neumann trace on the algebra, and $\chi(\mathcal{M}/\Lambda)$ is the Euler characteristic of the quotient space.

3. THE HCEM THEOREM

Theorem 3.1 (Hyper-Cohomological Entropy Manifold Theorem). *If $\mathcal{M}$ possesses strictly negative sectional curvature ($K < -1$) and Λ is a word-hyperbolic group, the problem of determining a specific representation $\rho : \Lambda \rightarrow U(\mathcal{A})$ from the partial observation of the entropy spectrum $\{S_\tau^{(k)}\}_{k=0}^n$ is computationally equivalent to finding the shortest vector in an infinite-dimensional*

non-commutative lattice. This is bounded below by complexity $\mathcal{O}(2^{c \cdot \text{Vol}(\mathcal{M}/\Lambda)})$, independent of quantum oracle access.

4. PROOF OF CERTAINTY AND SPECTRAL ASYMPTOTICS

Proof. Assume a quantum adversary possesses an oracle capable of executing the Quantum Fourier Transform (QFT). The QFT fundamentally requires an underlying abelian structure (or a solvable group structure) to extract periodicity via phase estimation.

By constraining Λ to be word-hyperbolic and $\mathcal{M}$ to exhibit strictly negative curvature, the geodesic flow on the unit tangent bundle $T^1(\mathcal{M}/\Lambda)$ is strongly ergodic (specifically, Anosov). Consequently, the spectrum of the twisted Laplacian Δ_k exhibits quantum chaos, conforming to Random Matrix Theory statistics (Gaussian Orthogonal Ensemble in the non-commutative limit).

We analyze the approximation of the spectrum. By the Selberg Trace Formula for non-commutative coefficients, the length spectrum of closed geodesics determines the eigenvalue spectrum. However, because Λ is non-abelian and hyperbolic, the growth rate of closed geodesics of length $\leq L$ is exponential:

$$(2) \quad N(L) \sim \frac{e^{hL}}{hL}$$

where h is the topological entropy.

Therefore, mapping the continuous entropy signature S_τ back to the discrete generator ρ necessitates searching an exponentially dense phase space with zero global periodicity. The mapping is thus a proven one-way function immune to quantum period-finding algorithms. $\square$

5. CONCLUSION ON GLOBAL NOVELTY

The introduction of the twisted Hodge-Laplacian spectrum as a public-key analogue across a non-commutative algebra creates a continuous-discrete hybrid manifold. This structure eliminates the vulnerabilities inherent in commutative lattices and isolated topological invariants, establishing a provable boundary against quantum cryptanalysis.

INDEPENDENT RESEARCHER, SOUTH SULAWESI, INDONESIA
Email address: aidilamrym@gmail.com